

NEXUSCALIBER // ANPR TRAJECTORY & ROUTE-ANOMALY ENGINE

Complete System Architecture, Technical Specification & Operations Manual (A to Z)

Project Code:	BEL PS 26127
Target System:	City-Wide ANPR Surveillance Network & ICCC Chennai
Document Version:	3.7.0 (Production Release)
Core Capabilities:	3-Factor Identity Fusion, Corridor Kinematics, Live Fullscreen Simulation, Edge RTSP Ingest
Classification:	Law Enforcement Technical Standard

Executive Summary: NexusCaliber delivers an end-to-end, high-concurrency surveillance engine engineered to ingest high-frequency optical vehicle sightings across municipal camera networks, reconstruct suspect trajectories with optical character ambiguity correction, calculate physical transit plausibility, detect multi-vehicle convoy formations, and present a cinema-grade command-center interface with real-time telemetry HUD.

1. System Topology & End-to-End Pipeline

The NexusCaliber platform is structured into three primary architectural tiers: **(1) Edge Vision Ingestion**, **(2) Core Mathematical Engine**, and **(3) Command & Control Frontend**.

Layer	Module	Primary Functionality
Edge Ingestion	camera_ingest.py ingest_service.py	RTSP stream capture, EasyOCR extraction, Indian plate grammar autocorrection, HSV vehicle color estimation, LRU deduplication.
Core Storage	database.py anpr.db (SQLite WAL)	Persistent relational storage with indexed queries for sightings, cameras, watchlists, corridor baselines, and supervisor audit logs.
Analytical Engine	match.py baseline.py	3-factor identity-fusion scoring, Haversine velocity computation, statistical corridor speed baseline modeling, convoy clustering, cloning detection.
REST API	main.py (FastAPI / Uvicorn)	Asynchronous high-concurrency API server with role gating (Operator vs Supervisor), anti-caching middleware, and RFC 4180 CSV export.
C2 Frontend	index.html, style.css app.js, views/*.js	Zero-dependency tactical dashboard, Fullscreen Cinema Map, Video-Player Playback toolbar, Real-Time Bottom Telemetry HUD, Web Audio alerts.

2. Mathematical Identity Fusion Engine

Correlating vehicle sightings across non-adjacent municipal camera nodes requires robust multi-modal weighting. NexusCaliber calculates a composite score $S_{composite} \in [0, 1]$ using three independent factors:

Composite Formula:

$$S_{composite} = (0.50 \times S_{plate}) + (0.30 \times S_{visual}) + (0.20 \times S_{transit})$$

Factor 1: Normalized Levenshtein Plate Similarity (Weight: 50%)

Measures edit distance between optical OCR readings, accounting for character substitutions. Score = $1.0 - (\text{LevenshteinDistance}(P1, P2) / \max(\text{len}(P1), \text{len}(P2)))$.

Factor 2: Visual Embedding & Feature Cosine Similarity (Weight: 30%)

Evaluates 16-dimensional visual feature vectors and vehicle color/type classifications using cosine similarity: Score = $(v1 \cdot v2) / (\|v1\| \times \|v2\|)$.

Factor 3: Physical Transit Kinematics Plausibility (Weight: 20%)

Calculates the Great-Circle Haversine distance (Δd) between camera GPS coordinates and elapsed time (Δt) to find velocity ($v = \Delta d / \Delta t$). Velocities within normal traffic speeds score 1.0; impossible speeds (>130 km/h) penalize the score and trigger route-anomaly or plate cloning alerts.

3. Statistical Anomaly & Convoy Detection

The engine maintains dynamically computed historical baselines for all camera corridors. Anomaly detection operates across four major categories:

Alert Type	Severity	Trigger Condition	Enforcement Action
SPEED_ANOMALY	MEDIUM	Velocity $v > \mu_v + 2.5\sigma_v$ on designated corridor.	Logged in Speed Violations leaderboard.
ROUTE_DEVIATION	MEDIUM	Transit delay $\Delta t > \mu_t + 3.0\sigma_t$ between adjacent nodes.	Flags vehicle for detour / off-grid analysis.
IMPOSSIBLE_TRANSIT	CRITICAL	Velocity $v > 180$ km/h between two distinct cameras.	Immediate Clone Plate alert & audio alarm.
BLACKLIST_HIT	CRITICAL	Sighting of plate registered on active police watchlist.	Critical toast notification, audio beeps.
CONVOY_DETECTED	HIGH	3+ vehicles detected traveling in formation across 2+ nodes within 10 min.	Coordinated transit tracking & convoy alert.

4. Tactical Operator Frontend & Cinema Simulation

The user interface is optimized for high-stress surveillance operations, featuring six primary views:

View	Key Capabilities & Features
1. Trajectory Search	Full-width 520px GIS map, Video-Player Playback toolbar (Play/Pause, Step Prev/Next, Scrubber Timeline, Speed Multiplier), Fullscreen Cinema Mode (100vw x 100vh), Live Bottom Telemetry HUD, and court-admissible PDF export.
2. Sighting Heatmap	Kernel density visualization powered by Leaflet.heat with dynamic zone filters (Central, North, South, West) and time windows (1h, 6h, 24h, 7d).
3. Blacklist Watchlist	Real-time hotlist verification, supervisor-gated plate registration with offense categories (Stolen, Wanted, Impound), and one-click removal.
4. Incident Alerts Triage	Severity filtering (Critical, High, Medium), alert acknowledgment workflow tagging officer Badge ID, 1-click trajectory jump, supervisor audit trail, CSV export.
5. Traffic Trends	Corridor hourly throughput bar chart, peak volume indicators, Speed Violations leaderboard, and Congestion Index.
6. Camera Health Grid	8-node hardware health monitor (ONLINE/DEMO/OFFLINE), Quad CCTV Live Stream matrix simulation, and camera registry CSV export.

Live Bottom Telemetry HUD ('Flight Recorder')

Docked at the bottom of the map view, this semi-transparent tactical HUD streams real-time telemetry as the simulated vehicle travels from point to point:

- **Active Corridor:** Origin Hop → Current Hop (e.g. CAM-ANNA-001 → CAM-TNAGAR-002)
- **Target Plate & Vehicle:** [CAR - WHITE] TN09CB1234
- **Speed & Distance:** 78.4 km/h | 3.82 km
- **Time & Delta:** 10:14:22 → 10:17:15 (+2m 53s)
- **Forensic Status:** High-contrast indicator for NORMAL transit or SPEED / ROUTE ANOMALY alerts.

5. Live Camera Network Deployment Guide

To connect physical CCTV IP cameras over RTSP into the NexusCaliber pipeline:

```
# Step 1: Install Python Computer Vision Dependencies
pip install opencv-python easyocr pyyaml

# Step 2: Configure backend/camera_config.yaml with RTSP URLs
cameras:
- id: 'CAM-ANNA-001'
  name: 'Anna Nagar Roundtana'
```

```
lat: 13.0850
lon: 80.2100
rtsp_url: 'rtsp://admin:Pass123@192.168.1.101:554/live'
enabled: true

# Step 3: Launch Watchdog Ingest Supervisor
py -3.13 backend/ingest_service.py
```

6. Quality Assurance & System Health

The automated Playwright test suite (run_verification.py) validates all critical functional paths across all 6 views with **zero console errors** and **zero uncaught exceptions**.

Test Component	Scope Tested	Status
View 1 Trajectory	Query execution, map polyline, video player controls, telemetry HUD	PASSED
View 2 Heatmap	Kernel density layer, zone filters, radius/blur sliders	PASSED
View 3 Blacklist	Watchlist lookup, supervisor registration, duplicate check, deletion	PASSED
View 4 Alerts & Audit	Severity filters, alert ACK, role gating (Operator vs Supervisor)	PASSED
View 5 Traffic Trends	Hourly volume bar chart, speed violation ranking table	PASSED
Audio & Toast System	Synthesizer beep generator, mute toggle, toast triggers	PASSED
Edge Cases	Nonexistent plates, clean watchlist checks, malformed payloads	PASSED
Browser Stability	Zero console errors, zero uncaught exceptions, static viewport	PASSED

7. SIH 2026 Problem Statement 127 (BEL) Alignment

Problem Statement Title: City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics
Organization: Bharat Electronics Limited (BEL) | **PS Code:** SIH26127 / PS 127

NexusCaliber matches 100% of the BEL SIH #127 challenge scope while introducing five proprietary innovations:

SIH PS #127 Requirement	NexusCaliber Implementation	Evaluation Advantage
Multi-Camera Ingestion	Threaded RTSP pipeline with buffer zeroing & auto-reconnection.	Eliminates network streaming latency & handles packet drops.
Cross-Camera Vehicle Re-ID	3-Factor Identity Fusion (OCR Levenshtein + Visual Vector + Haversine Physics).	Accurate cross-node Re-ID even with partially obscured plates.
Indian Plate Grammar Repair	Deterministic RTO slot regex autocorrection (e.g. TN 09 CB 1234).	Corrects '0' vs 'O', '8' vs 'B', '1' vs 'l' misreads at edge.
Trajectory Tracking & Playback	Leaflet GIS map, 520px full width, Video-Player scrubber, Fullscreen Cinema Mode.	Inspect suspect transit like reviewing a flight video with zero page scroll.
Live Bottom Telemetry HUD	Semi-transparent tactical HUD ('Flight Recorder') docked on map.	Real-time streaming of active corridor, speed, distance, elapsed time, alert tags.
Route & Speed Anomalies	Historical corridor baseline modeling with IQR / Z-score deviations.	Autonomous speeding and detour detection without manual threshold setting.
Plate Cloning Detection	Kinematic velocity validator (impossible velocity $v > 180$ km/h).	Immediate CRITICAL alert for stolen or cloned vehicle plates.
Convoy Formation Detector	Clustering 3+ vehicles traversing 2+ sequential checkpoints within 10 min.	Identifies coordinated criminal, smuggling, or VIP convoy groups.
Traffic Flow Analytics	Hourly corridor throughput bar charts, Congestion Index, Speed Violations table.	City-wide bottleneck mitigation for ICCC municipal planners.
Law Enforcement Integrity	Officer Badge ID alert ACK logging, supervisor audit trail, CSV export.	Court-admissible forensic accountability for police operations.